

Assignment - 1

Group - 1



Q1 - Object Detection and Depth Estimation

Task A:

Record data and train Yolo model. Show model performance

Task B:

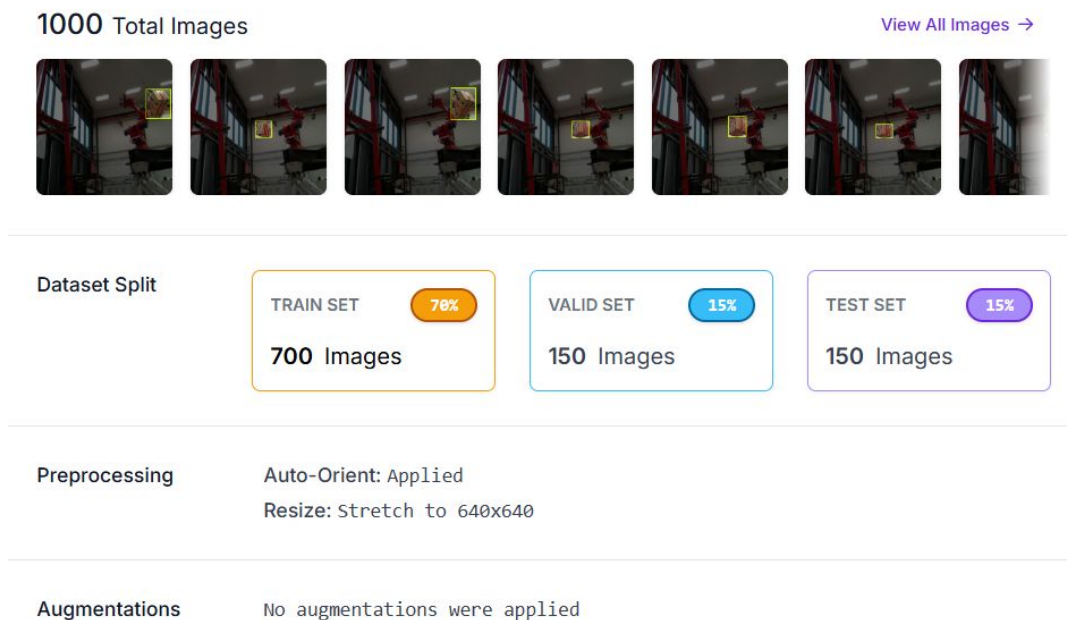
Display output from model

Task C:

Find depth and placement of box

Task A

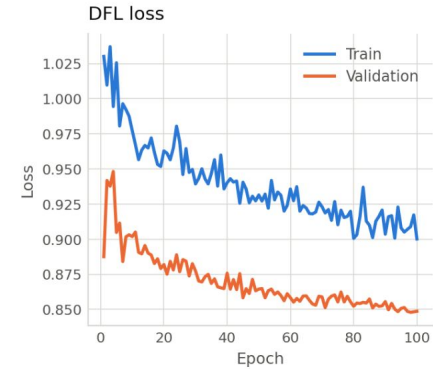
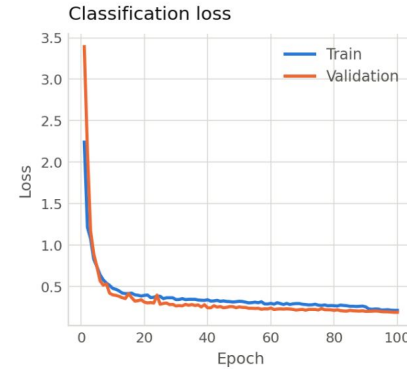
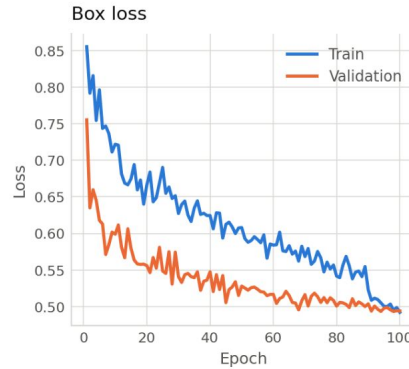
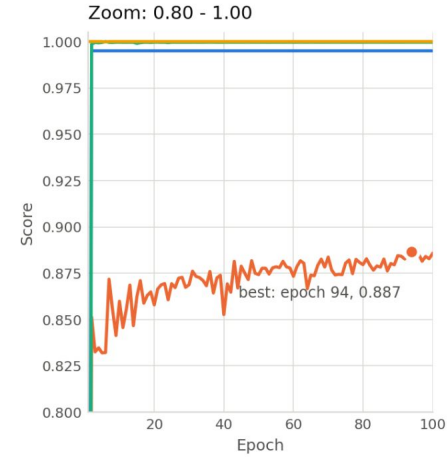
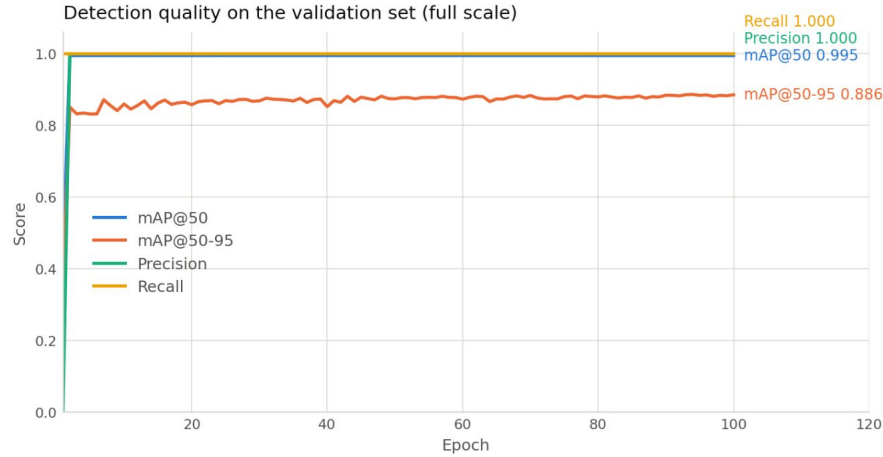
- Data set
 - 1000 images
 - 700 train
 - 150 test
 - 150 val
- Labeled using roboflow
- Stretched to 640x640
- No augmentation



YOLOv11n on the 'Load' dataset - 100 epochs

Task A

- Yolo v11 nano
 - 12 min to train
 - Only 1 class
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Task B

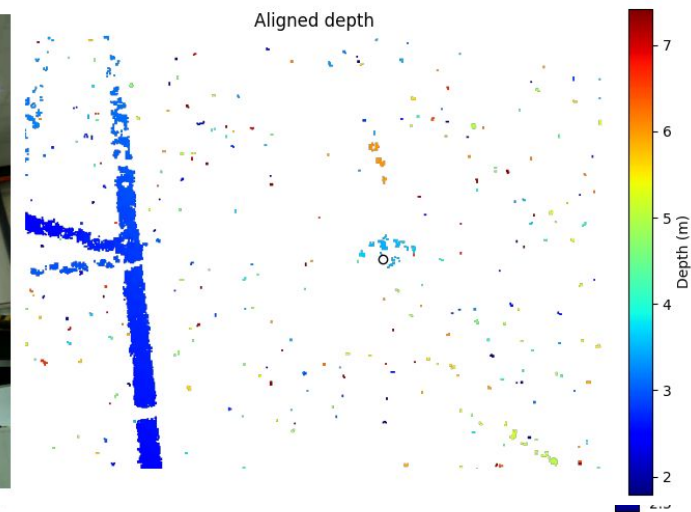
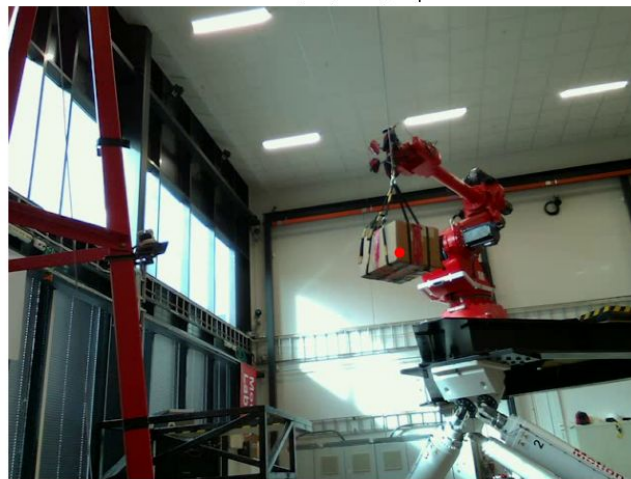
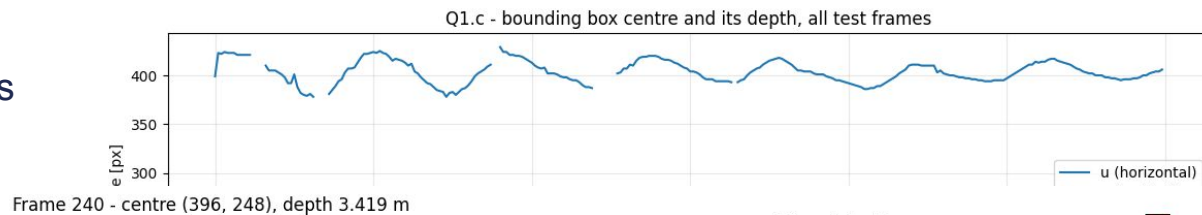
- Sample 5 images and run model
- Used on data the model has not seen before
- Location shown by pixel count

Test frame 261

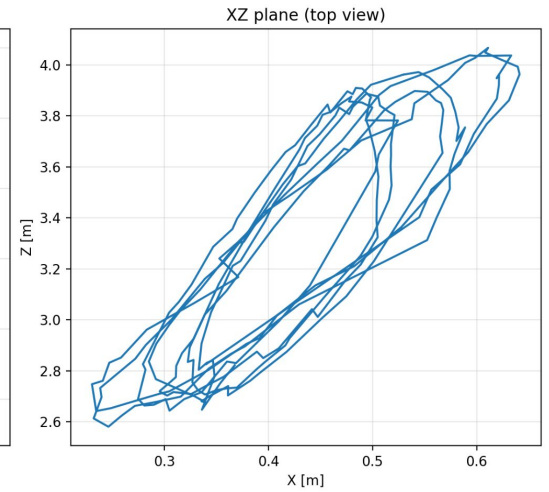
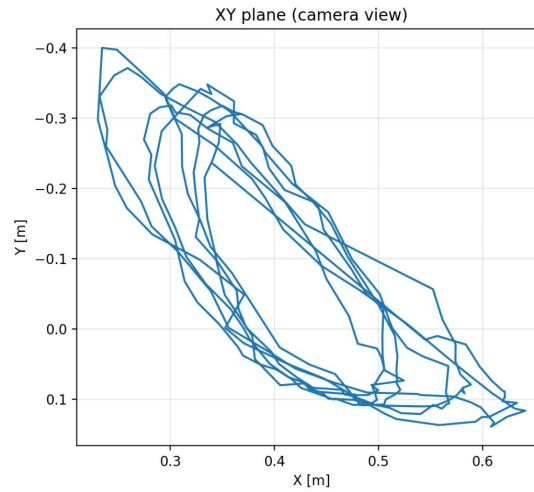


Task C

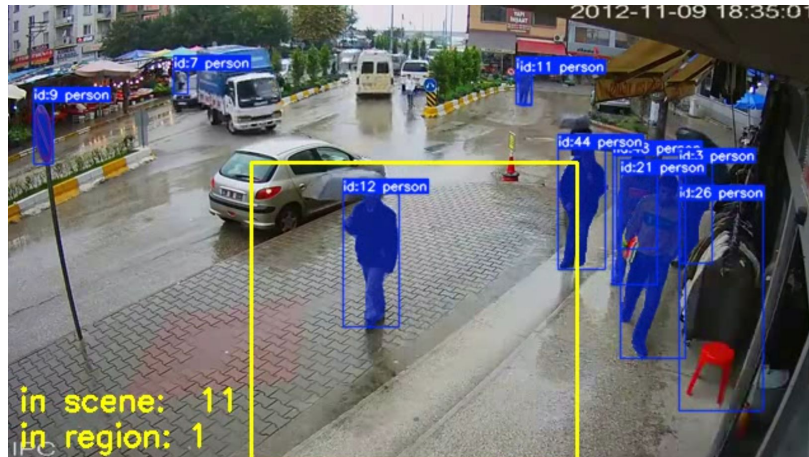
- Find the depth of all images
- Model loss



Q2 - Object Tracking

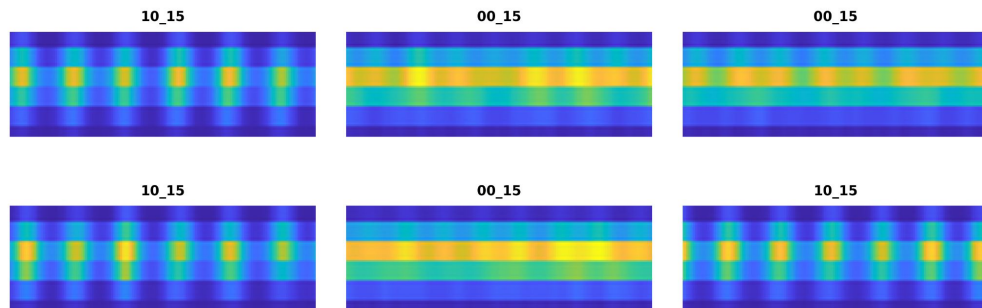


Q3 - Multi Object Segmentation, Tracking and Counting

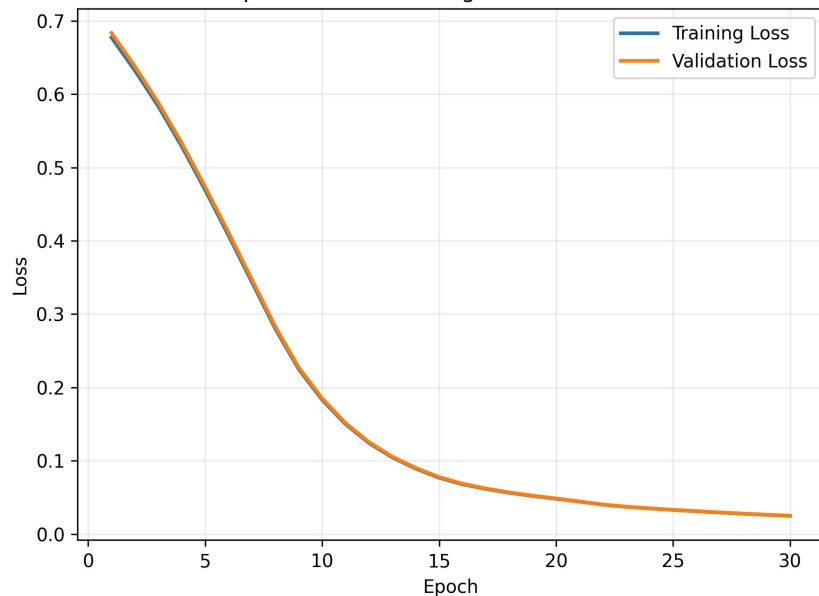


Q4 - Classification

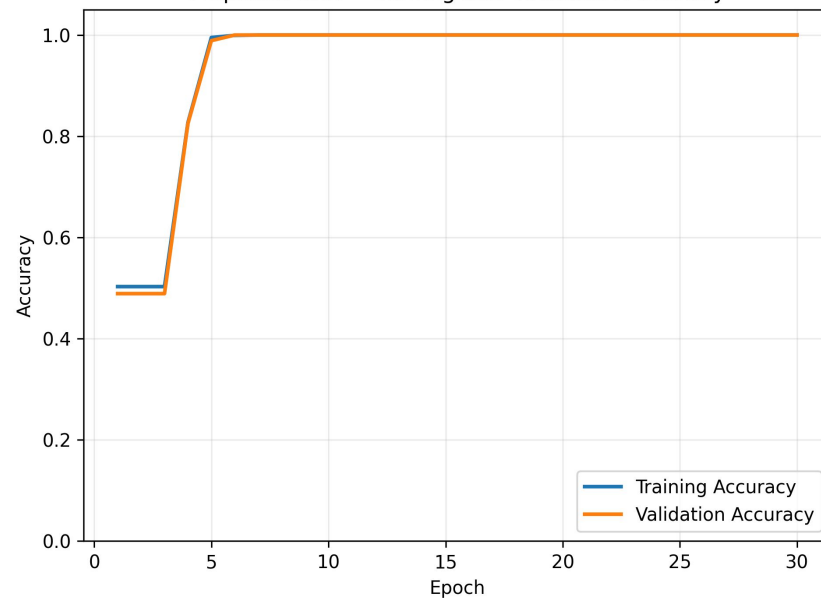
Motor Fault Dataset - Training Examples



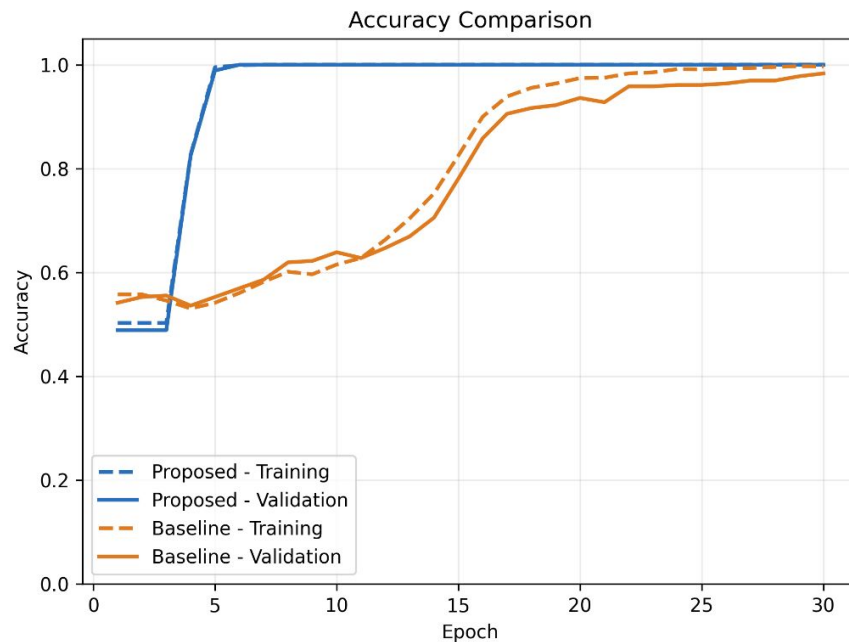
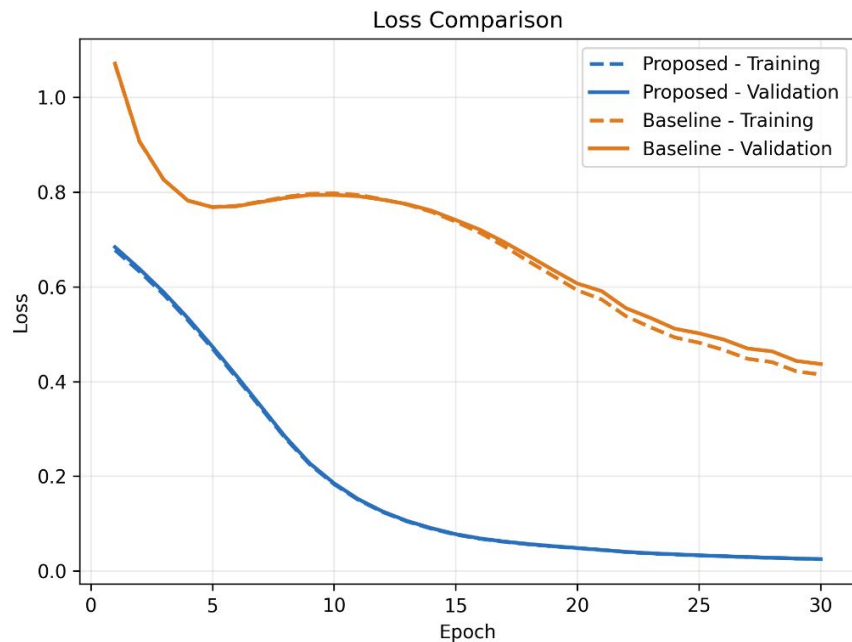
Proposed CNN - Training and Validation Loss



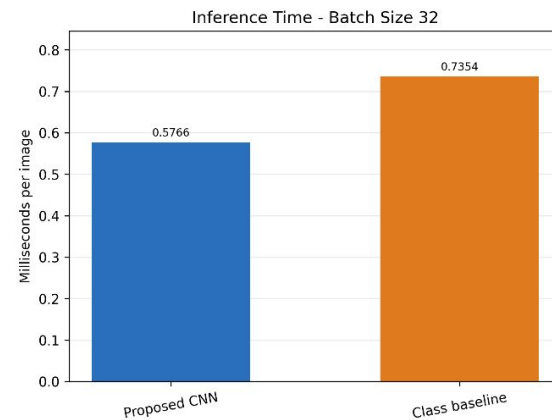
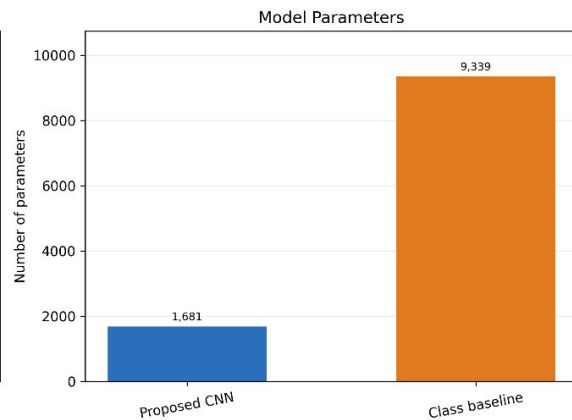
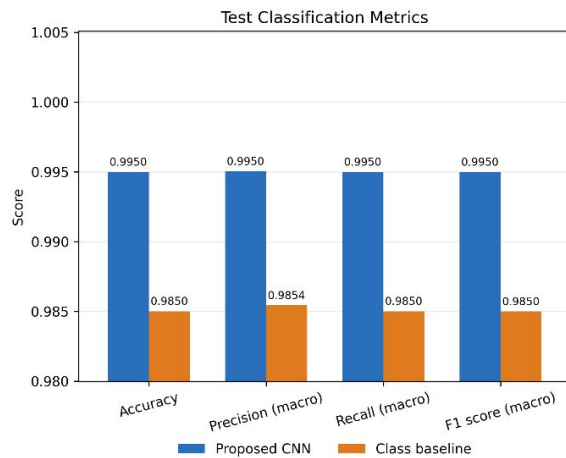
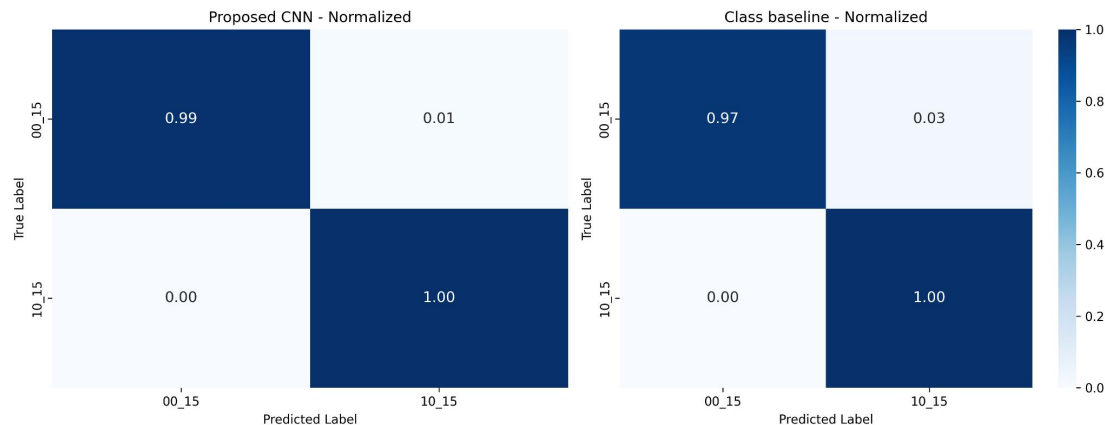
Proposed CNN - Training and Validation Accuracy



Q4 - Classification

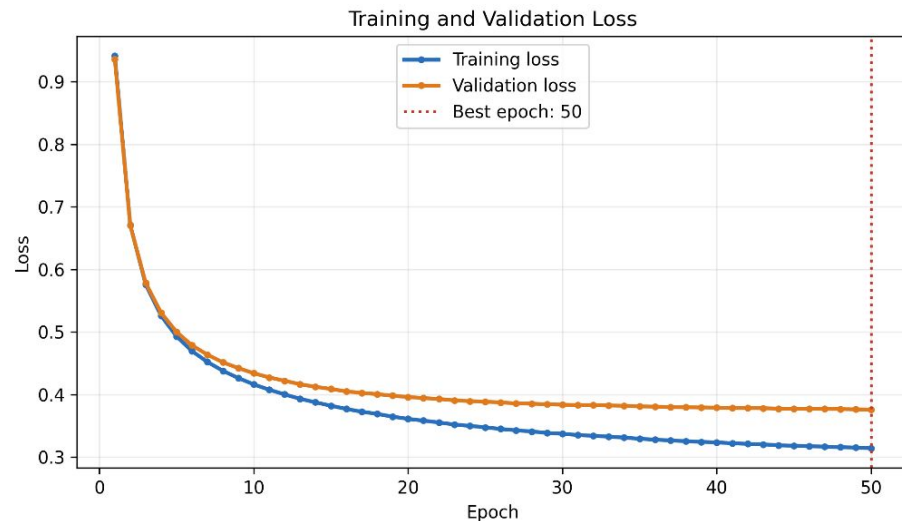
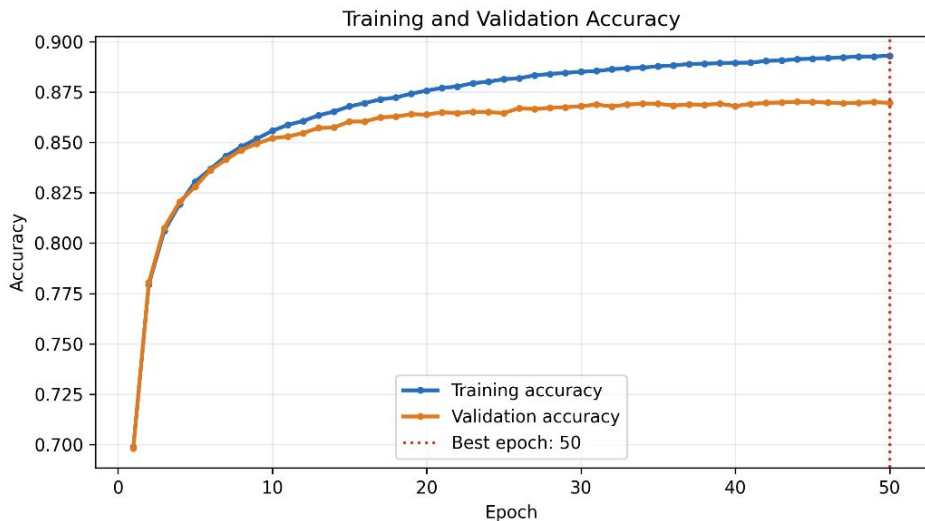


Q4 - Classification

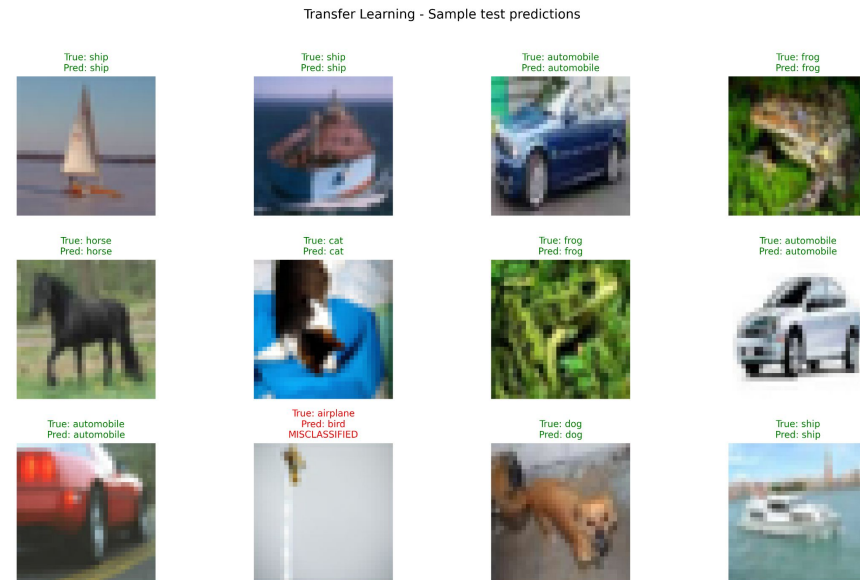
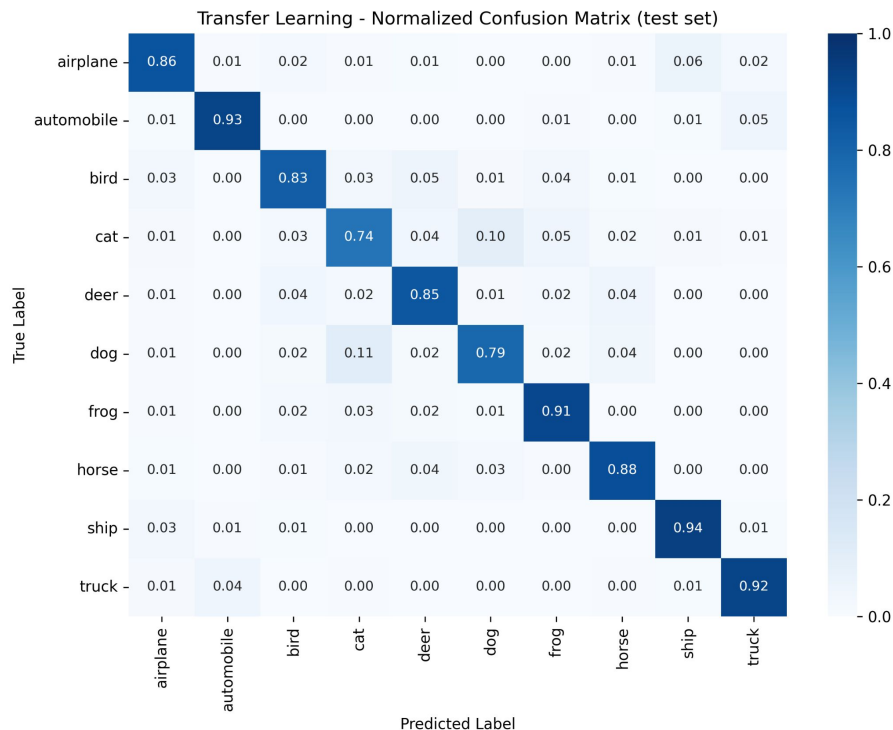


Q5 - Classification - Transfer Learning

Transfer Learning - Learning Curves
Best epoch by validation loss: 50 - validation accuracy: 0.8697 - validation loss: 0.3760

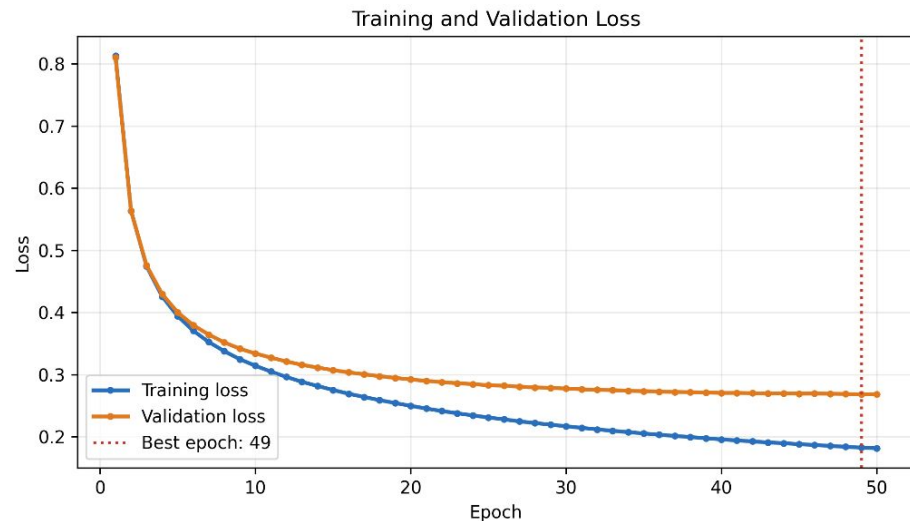
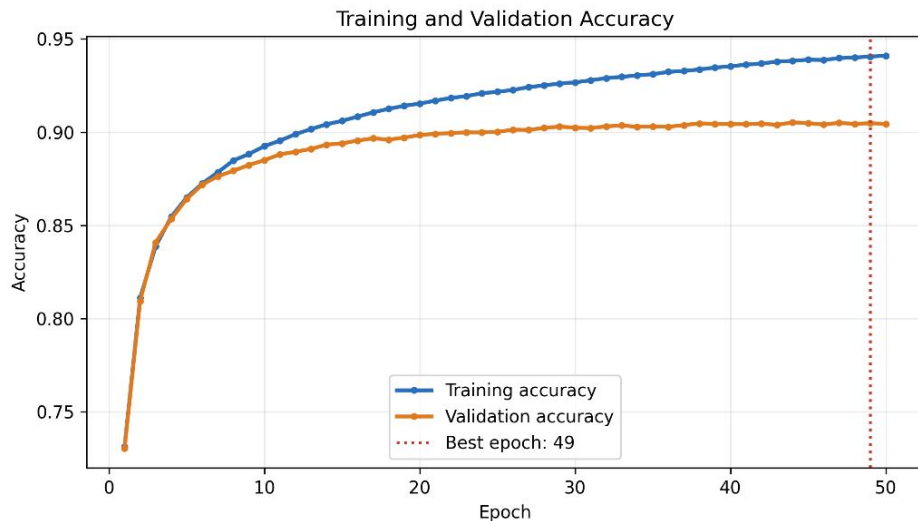


Q5 - Classification - Transfer Learning

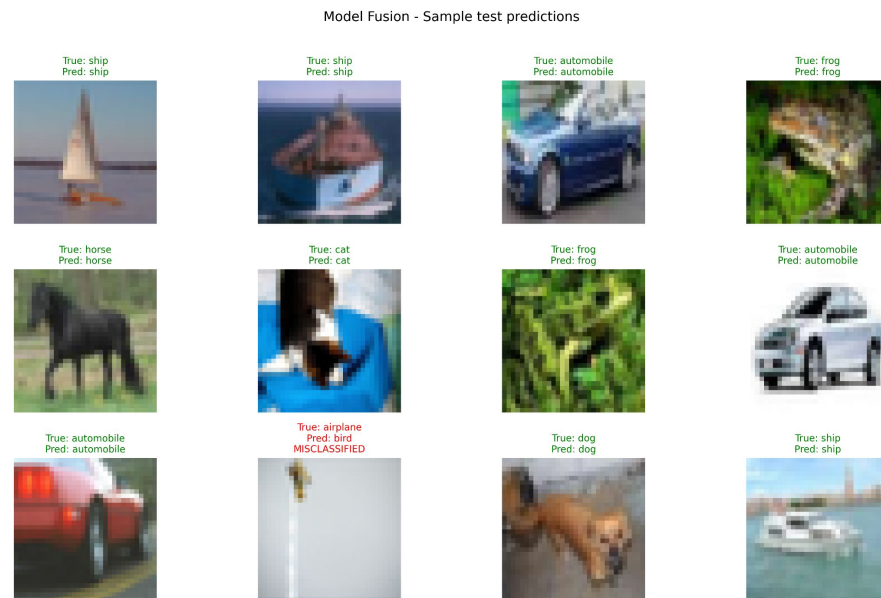
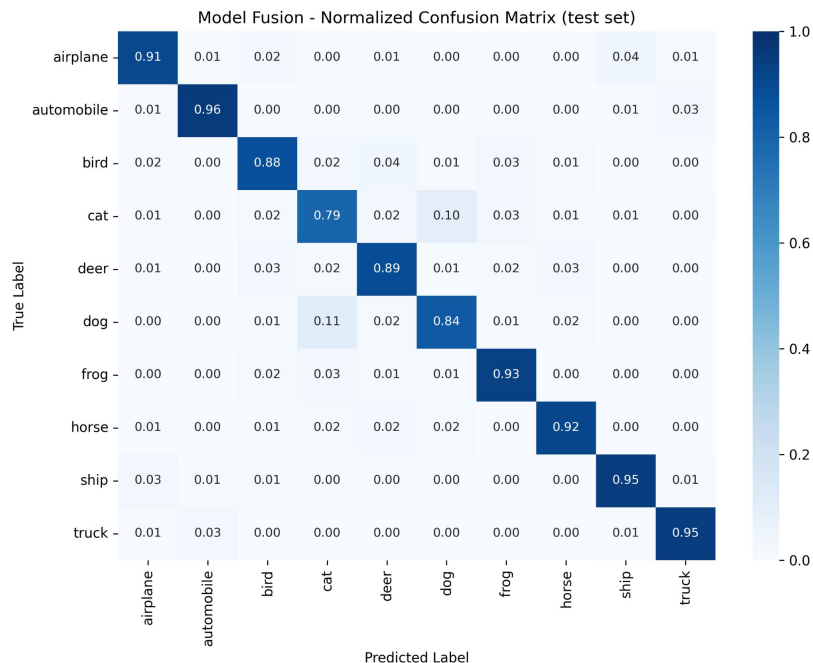


Q5 - Classification - Model Fusion

Model Fusion - Learning Curves
Best epoch by validation loss: 49 - validation accuracy: 0.9048 - validation loss: 0.2683

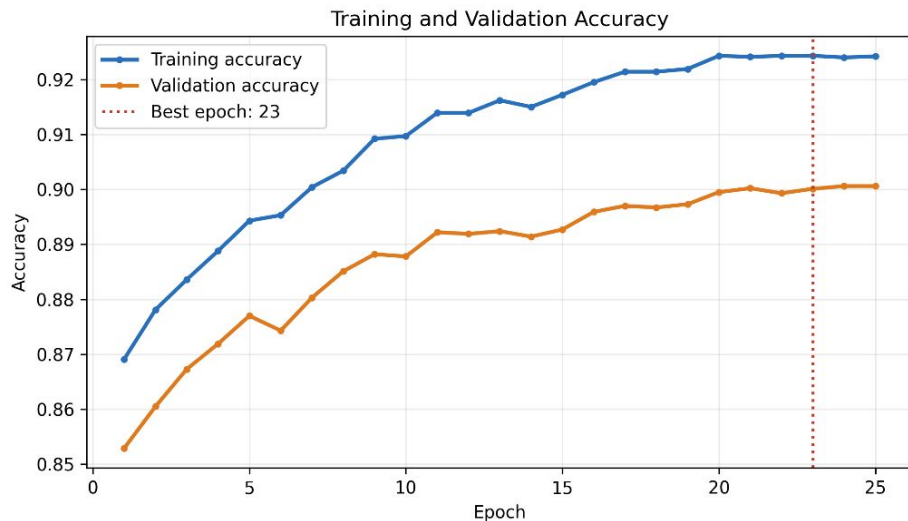


Q5 - Classification - Model Fusion



Q5 - Classification

Fine-tuning - Learning Curves
Best epoch by validation loss: 23 - validation accuracy: 0.9001 - validation loss: 0.3172

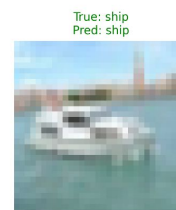
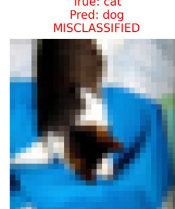
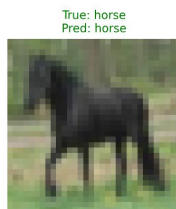
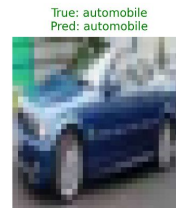


Q5 - Classification

Fine-tuning - Normalized Confusion Matrix (test set)

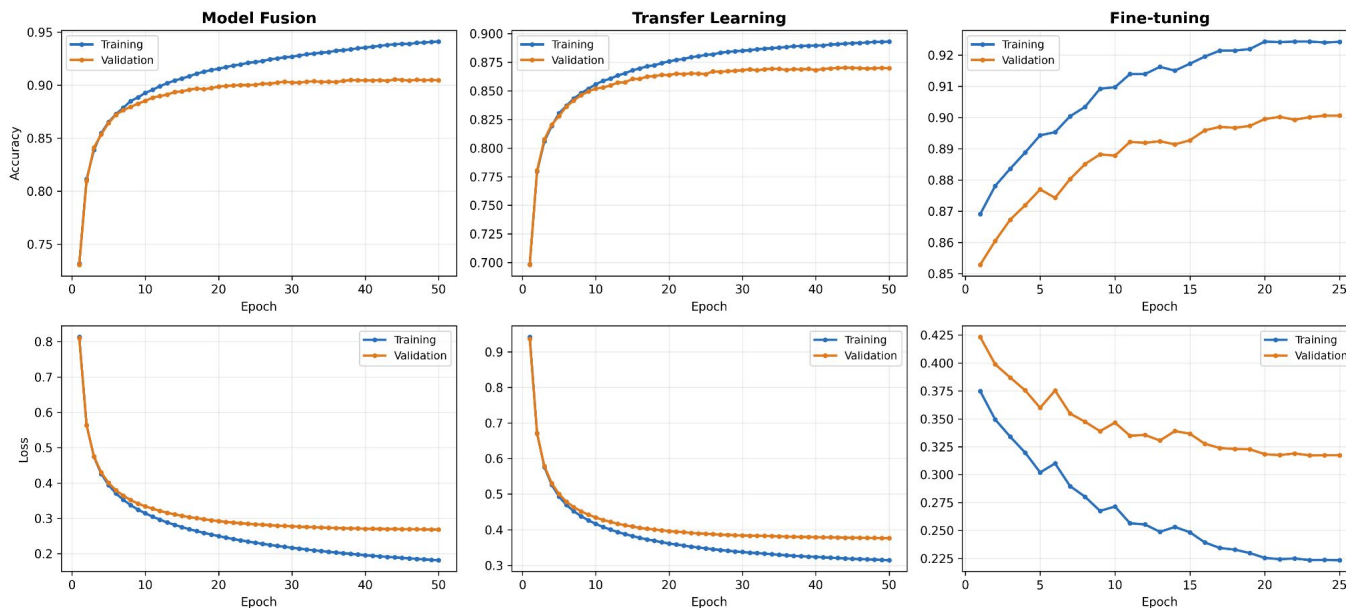
True Label \ Predicted Label	airplane	automobile	bird	cat	deer	dog	frog	horse	ship	truck
airplane	0.91	0.01	0.02	0.01	0.01	0.00	0.01	0.01	0.03	
automobile	0.00	0.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
bird	0.01	0.00	0.86	0.02	0.02	0.01	0.05	0.01	0.00	
cat	0.01	0.00	0.02	0.74	0.03	0.09	0.08	0.02	0.01	
deer	0.01	0.00	0.01	0.02	0.85	0.01	0.05	0.04	0.00	
dog	0.00	0.00	0.01	0.10	0.01	0.82	0.04	0.02	0.00	
frog	0.01	0.00	0.01	0.01	0.01	0.00	0.97	0.00	0.00	
horse	0.01	0.00	0.01	0.01	0.02	0.02	0.01	0.92	0.00	
ship	0.04	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.94	
truck	0.01	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01

Fine-tuning - Sample test predictions

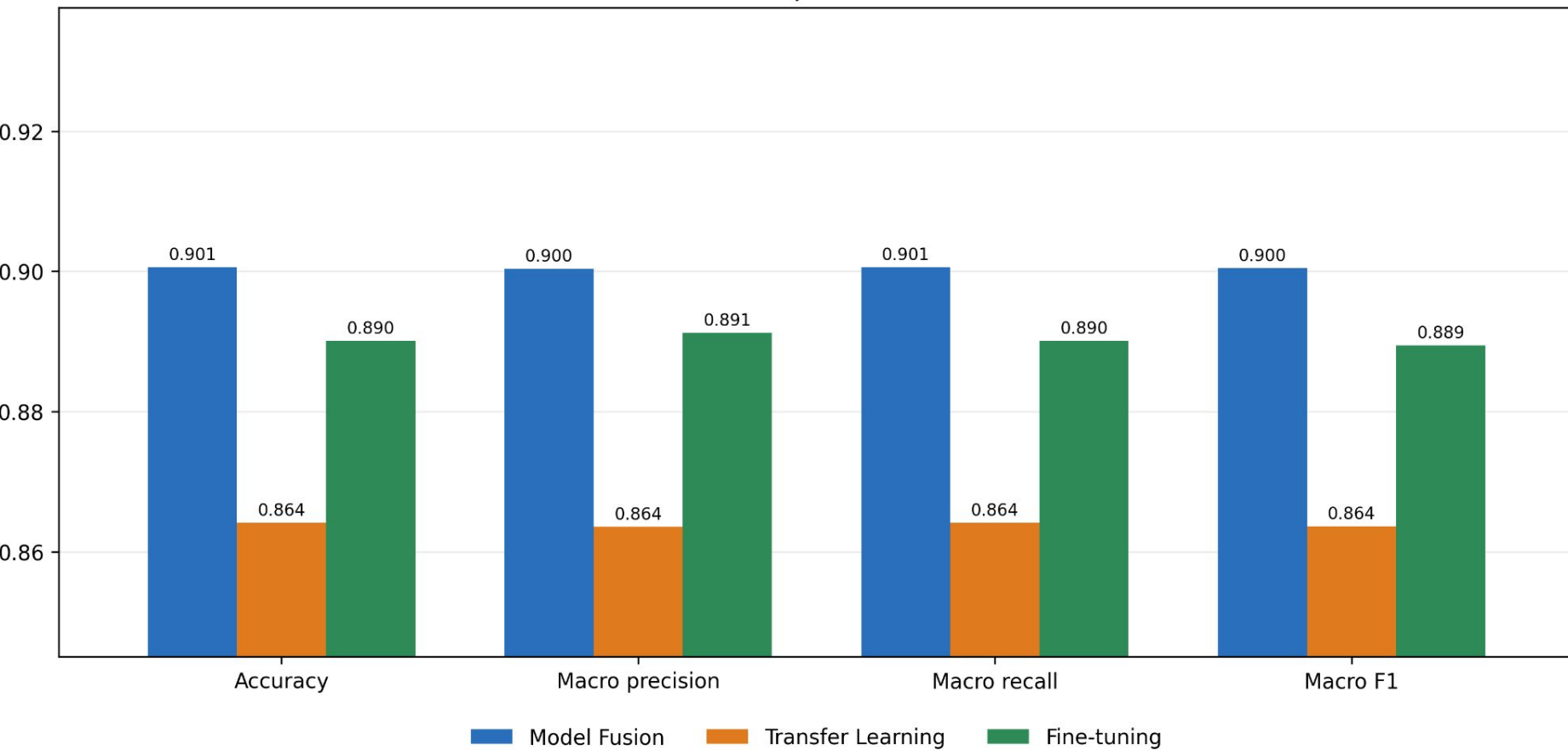


Q5 - Classification - Comparison

CIFAR-10 - learning curves of the three methods

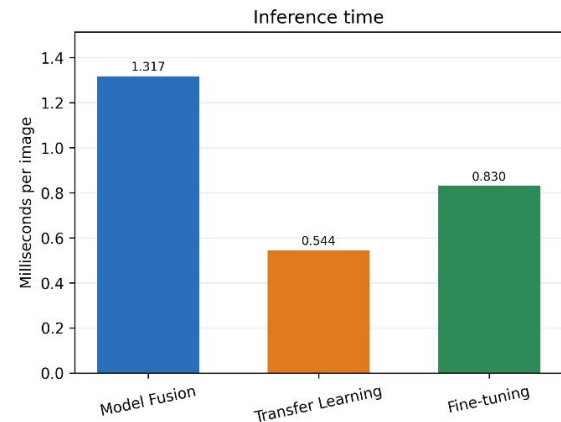
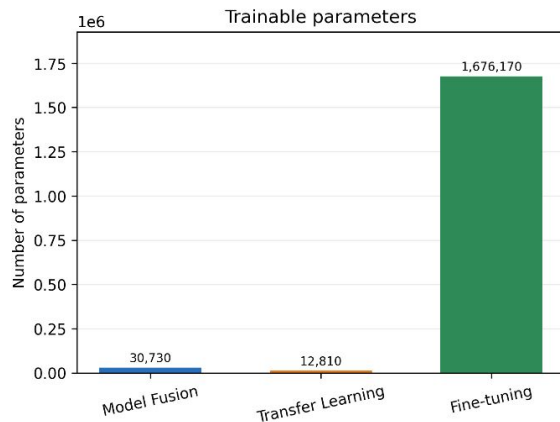
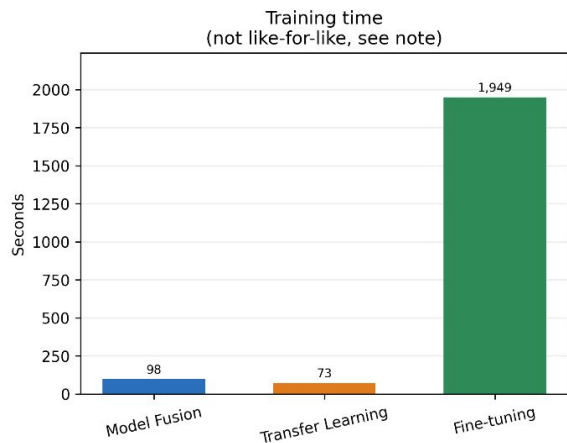


CIFAR-10 test-set KPI comparison of the three methods

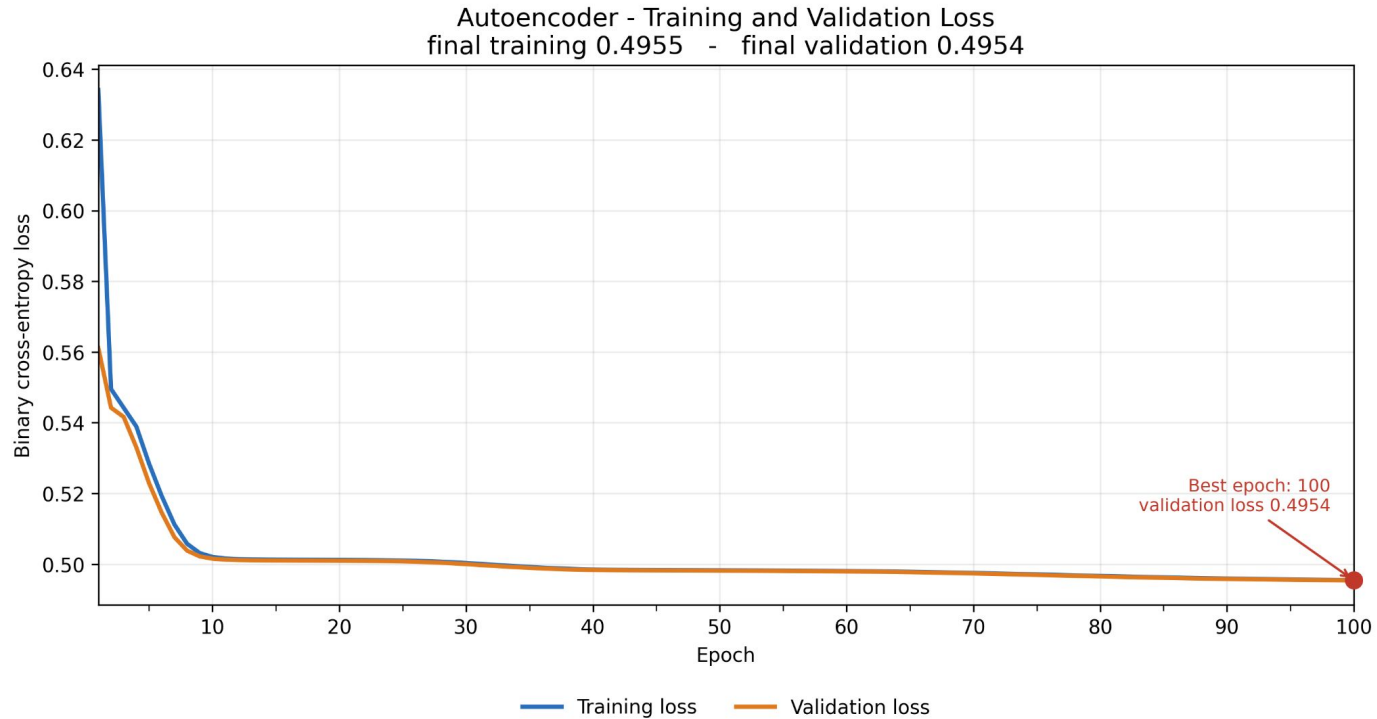


Q5 - Classification - Comparison

Cost of each method

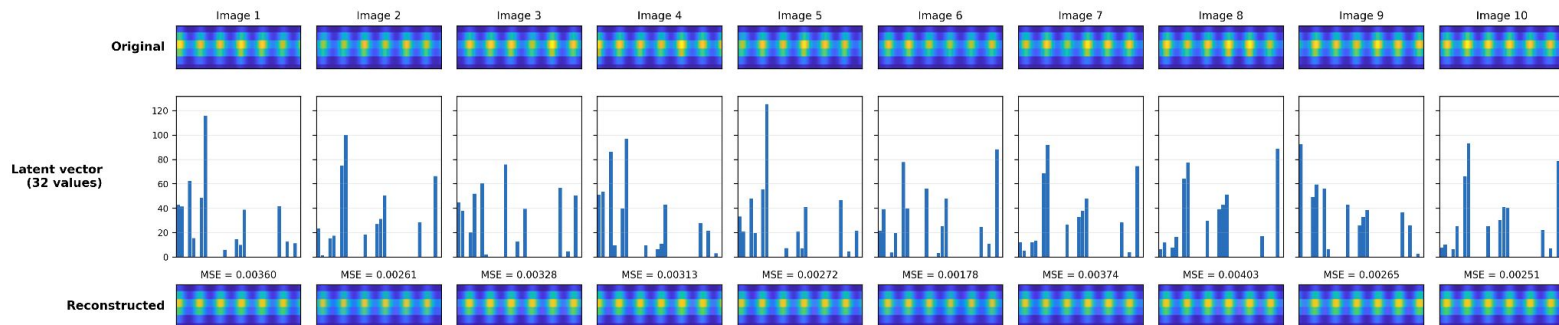
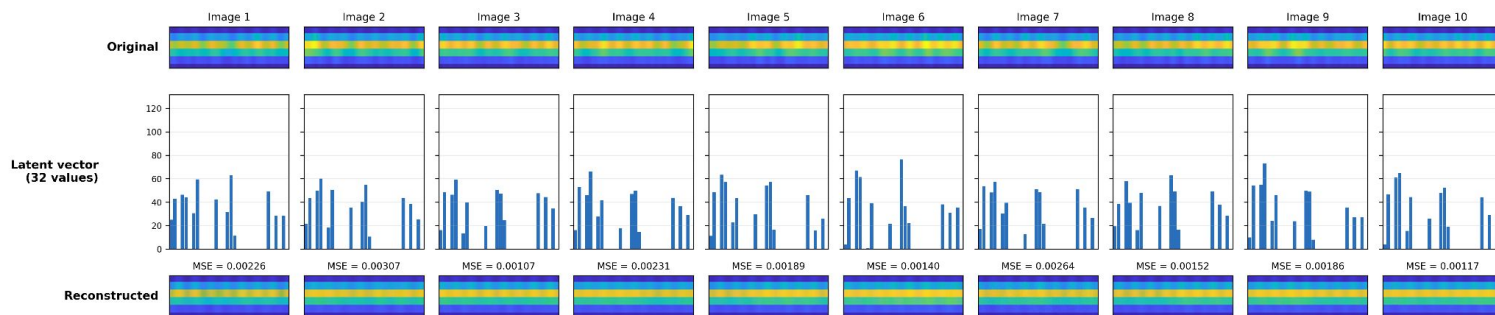


Q6 - Autoencoder



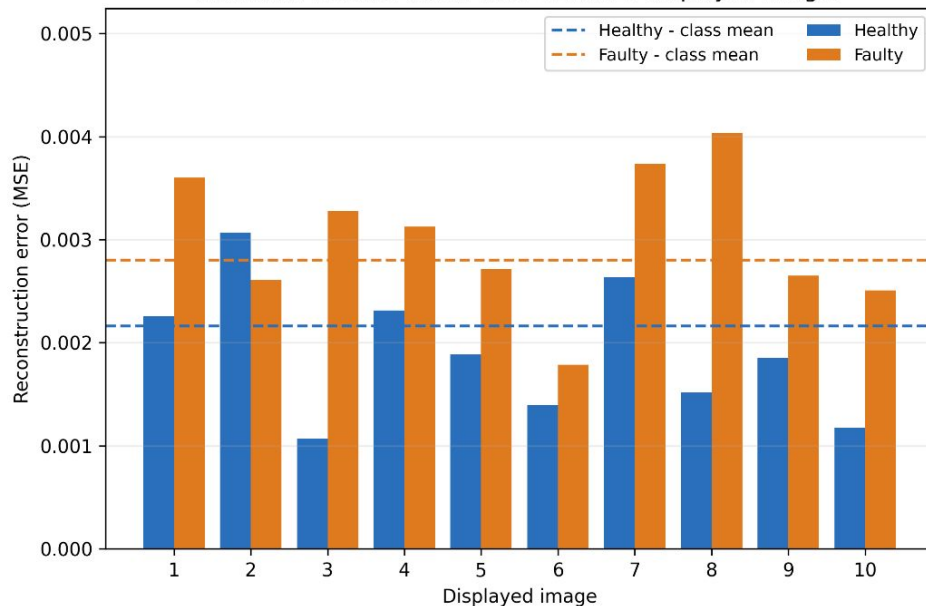
Q6 - Autoencoder

Autoencoder results - Class: Healthy



Q6 - Autoencoder

Individual reconstruction error of the 10 displayed images



Error distribution over the whole test set

